

Numerical PDEs Homework #0

For all problems asking for a graph: you do not need to write up the method you programmed. Including plots of your solution and convergence is sufficient for full credit. The ideal homework will have one, information-dense and properly labeled figure with a caption and a *brief* discussion. **You must type up your HW.** I don't care if it's word or Tex as long as it's well organized, easy to follow, and submitted as a pdf.

Problem 1: Finite differences Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

1. Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.

Solution:

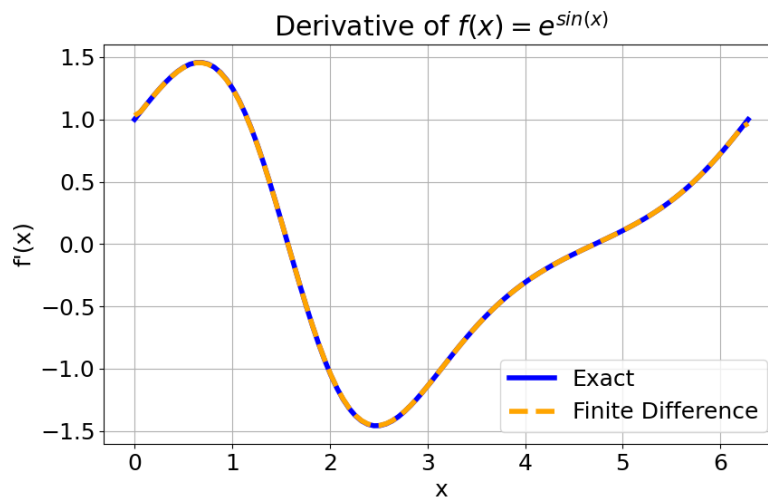


Figure 1: Comparison of the exact derivative of $f(x) = e^{\sin(x)}$ with the numerical finite-difference approximation, using second-order centered differences at interior points and first-order one-sided differences at the boundary points ($x = 0$ and $x = 2\pi$).

Figure 1 shows that the finite-difference approximation is visually indistinguishable from the exact derivative across the internal grid points. The end points slightly differ from the exact derivative. This is not surprising given that the endpoints use only first-order one-sided difference ($O(h)$ error), while the interior points use second-order centered difference ($O(h^2)$ error).

2. Show a relative error convergence plot by plotting the relative error on the y-axis and the

grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and 3/2 respectively, i.e. it scales with $\frac{1}{N}, \frac{1}{N^{3/2}}$. Extra Credit: The 3/2 slope is strange, why is that happening?

Solution:

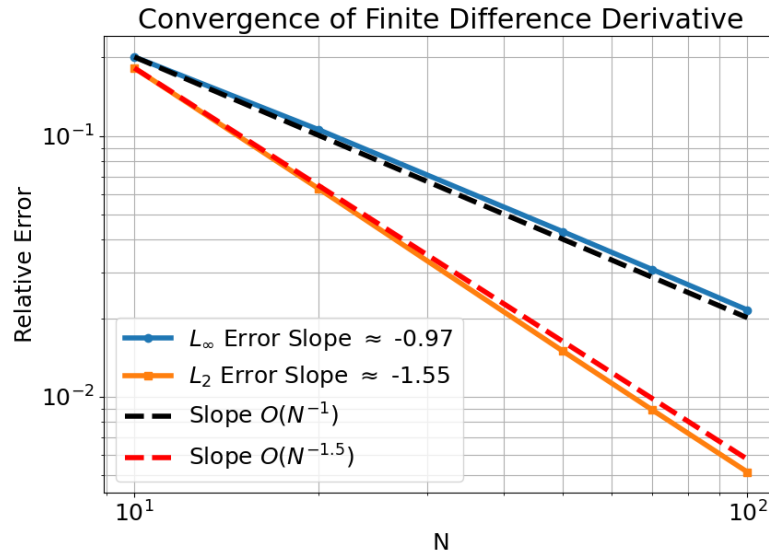


Figure 2: Convergence of Finite Difference First Derivative.

Figure 2 shows that both the L_∞ and L_2 relative errors decrease as N increases, confirming convergence of the finite-difference scheme. The L_∞ error decreases with a slope of approximately -0.97 , matching the expected first-order rate of $O(N^{-1})$, while the L_2 error decreases faster, with a slope of approximately -1.55 , close to the expected $O(N^{-\frac{3}{2}})$ rate.

Extra Credit:

The L_∞ error is set by the worst point, which is one of the two endpoints where the one-sided difference has $O(h)$ error, so $L_\infty \sim N^{-1}$. For L_2 , summing squared errors gives $2 \cdot O(h^2)$ from the two endpoints and $N \cdot O(h^4) = O(h^3)$ from the interior, so the endpoints dominate and $\|f' - f'_n\|_2 = O(h)$. The discrete 2-norm is an unweighted sum over N points, so $\|f'\|_2 \sim \sqrt{N}$ grows with the grid, while the error is concentrated in only two points whose count does not grow. Dividing gives

$$Err_2 \sim \frac{O(h)}{\sqrt{N}} = O(N^{-1}) \cdot O(N^{-1/2}) = O(N^{-3/2}).$$

If the norm were scaled by $\sqrt{h}$ (a proper approximation to the continuous L_2 norm), the numerator would be $\sqrt{h} \cdot O(h) = O(h^{3/2})$ and the denominator would be $O(1)$, giving the same rate, the $3/2$ reflects $O(h)$ errors living on a set of measure $O(h)$.

Problem 2: Second-order derivatives Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Solution:

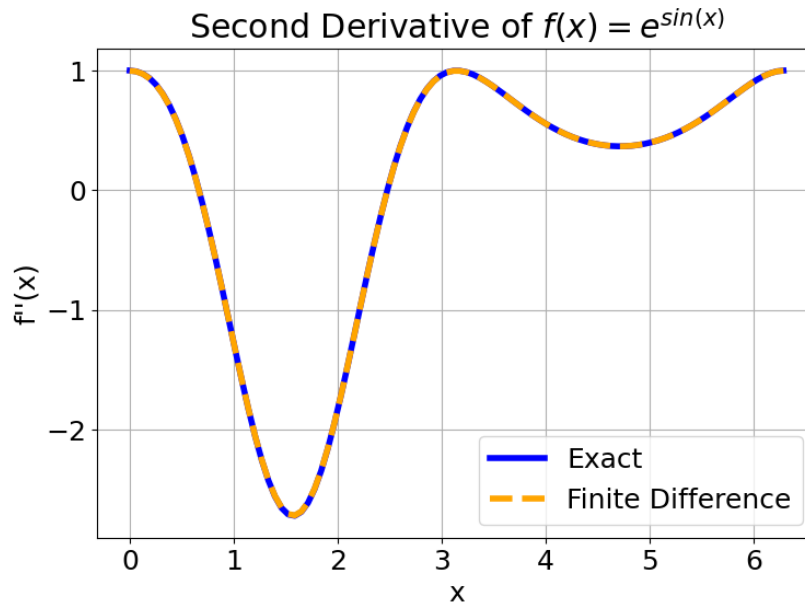


Figure 3: Comparison of the exact second derivative of $f(x) = e^{\sin(x)}$ with the numerical finite-difference approximation, using second-order centered differences at interior points and periodicity at the boundary points ($x = 0$ and $x = 2\pi$).

Figure 3 shows that all points including the endpoints are visually indistinguishable from the exact solution for the second derivative of the function.

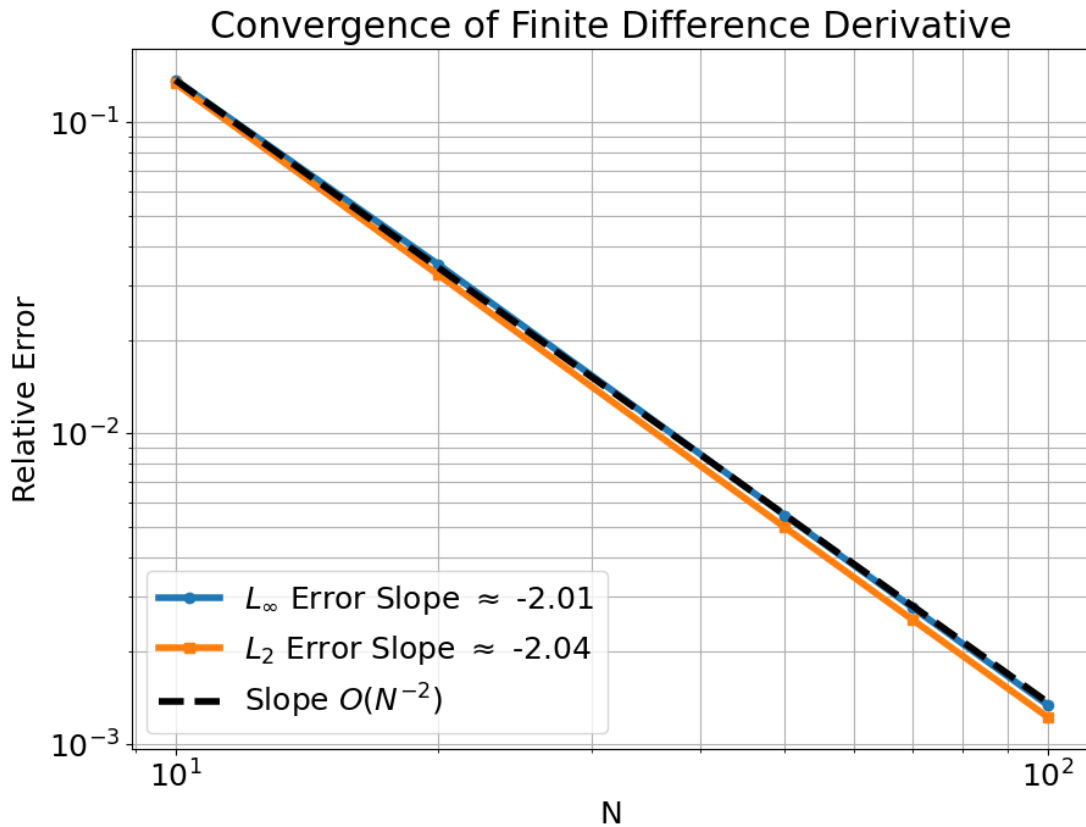


Figure 4: Relative error of the periodic finite-difference second derivative versus N (log-log), with reference slope N^{-2}

This shows that the slope of the maximum error for each N is decreasing as N increasing and the average error for each N is also decreasing as N increases.

Problem 3: ODE Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Solution:

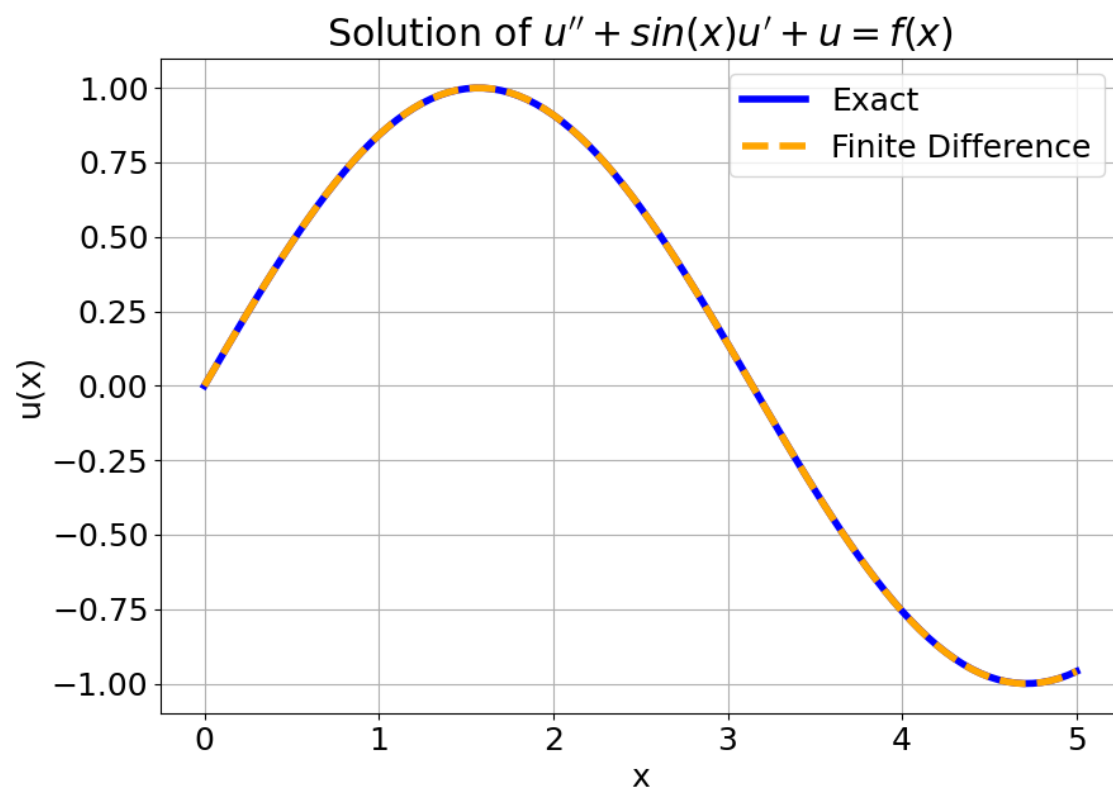


Figure 5: Solution to $u'' + \sin(x)u' + u = f(x)$, where $u = \sin(x)$ and $0 \leq x \leq 5$.

Figure 5 shows that all points including the endpoints are visually indistinguishable from the exact solution for the ODE.

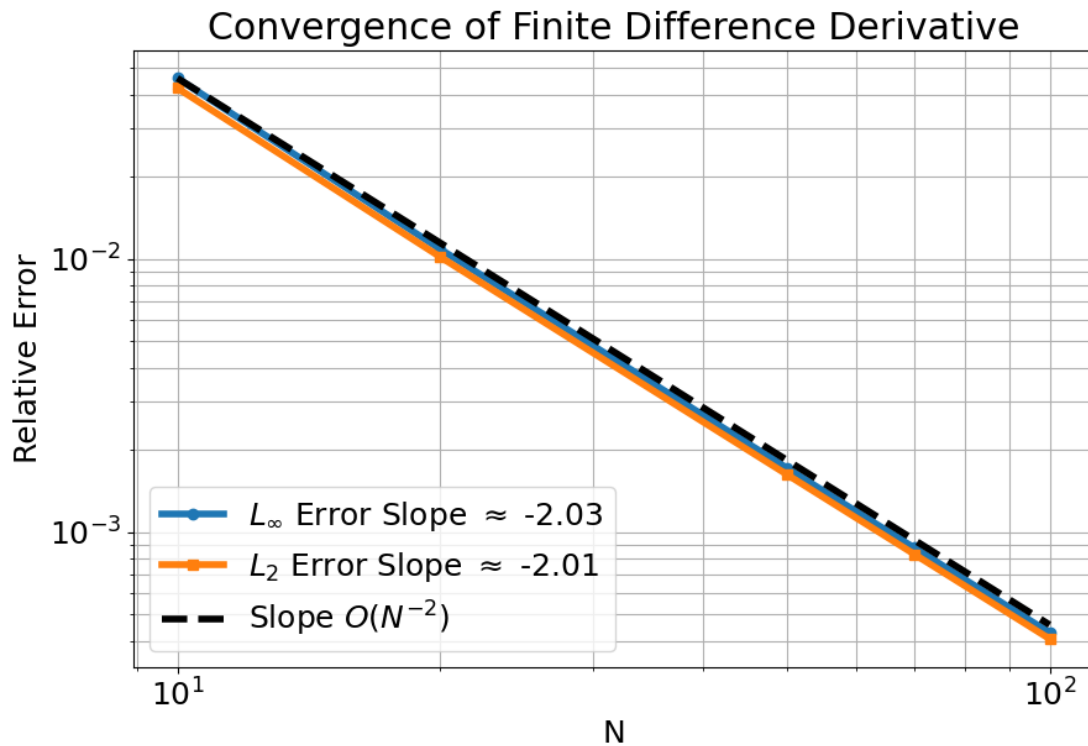


Figure 6: Convergence of Finite Difference ODE.

This shows that the slope of the maximum error for each N is decreasing as N increasing and the average error for each N is also decreasing as N increases.

Problem 4: Poisson Equation Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.

1. Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).

Solution:

Numerical Solution to Poisson Equation

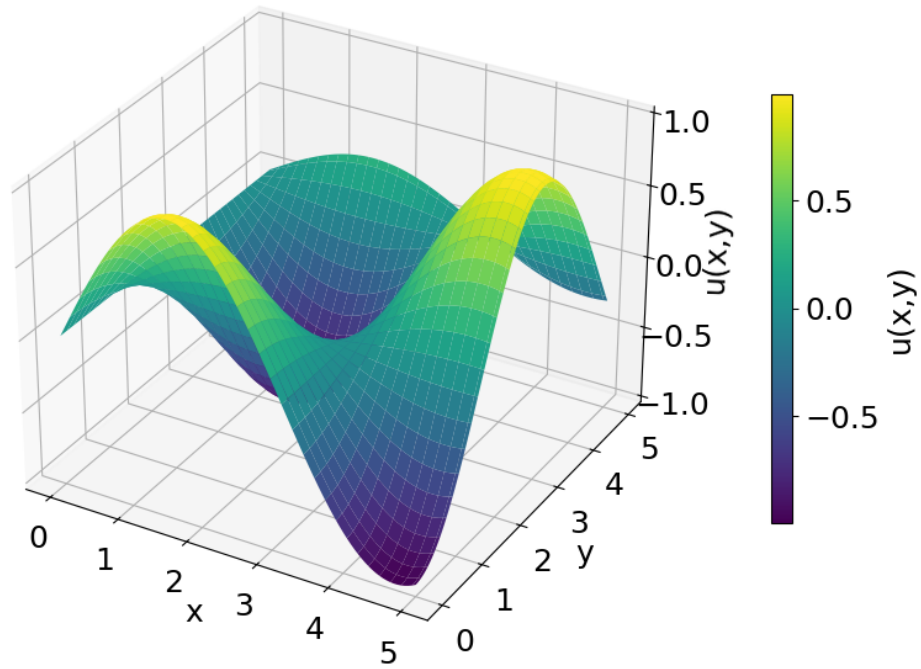


Figure 7: Numerical Solution to Poisson Equation for $u(x,y) = \sin(x)\cos(y)$

Figure 7 shows the numerical solution to the 2D Poisson equation with Dirichlet boundary conditions, computed on the domain $[0, 5] \times [0, 5]$. The solution exhibits the expected oscillatory pattern of $u(x,y) = \sin(x)\cos(y)$, with two peaks yellow, $u \approx 1$ and dark purple at $u \approx -1$. The solution stays bounded within $[-1, 1]$ as expected from the exact solution's range.

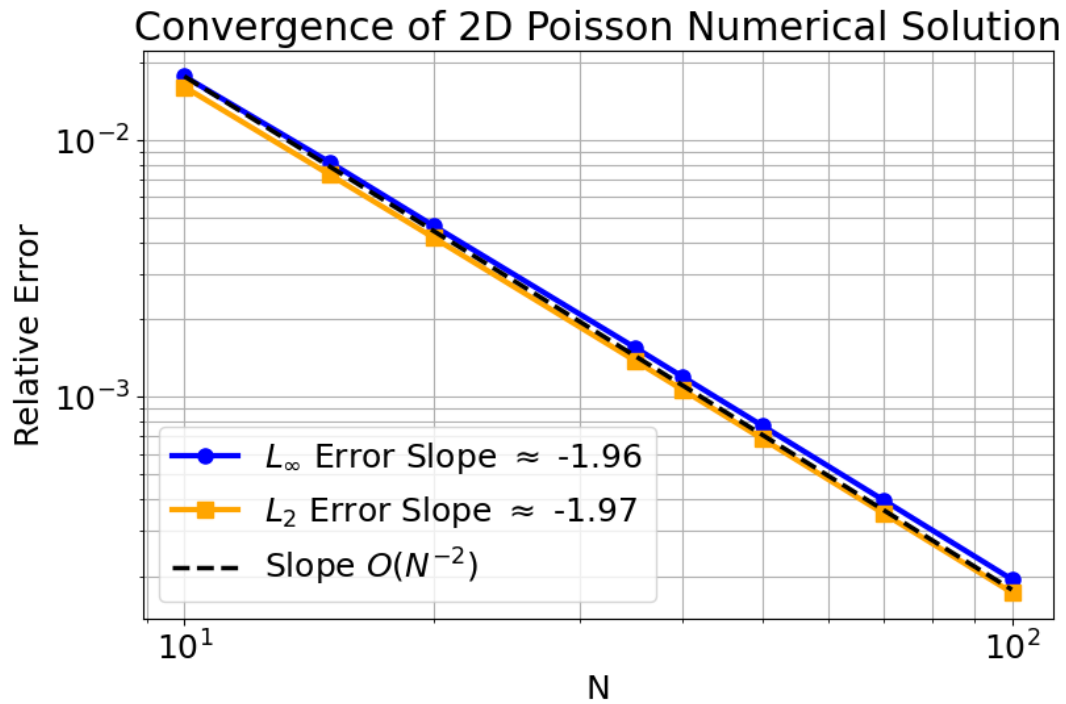


Figure 8: Caption

Figure 8 shows the relative error of the numerical solution to the 2D Poisson equation as a function, plotted on a log-log scale. Both the L_∞ and L_2 errors decrease steadily with increasing N .

2. Modify your code to apply Neumann conditions on all boundaries. **This won't work.** Why not? – both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

Solution: For the Neumann condition, we use the derivative of the test solution as the boundaries. The partial derivative of the test solution are

$$\frac{\partial u}{\partial x} = \cos(x) \cos(y) \quad \frac{\partial u}{\partial y} = -\sin(x) \sin(y)$$